

MSc Research Project/Internship/ Practicum

Introduction

Introduction Section

- **The introduction gives an overall review of the paper**
- **It works upon the principle of introducing the topic of the paper and setting it into a broad context, gradually narrowing down to a research problem, thesis and hypothesis.**
- **A good introduction explains how you mean to solve the research problem and creates ‘leads’ to make the reader want to delve further into your work.**
- **Introduction Section should present**
 - Background
 - Importance
 - Your research question and objectives
 - Limitations
 - Outline the structure of the report

Introduction - Background

- **The first task of the Introduction section is to set the scene, giving your paper a context and seeing how it fits in with previous research in the field. It can be based around a historical narrative, from the very first research in the field to the current day**
- **The background to the problem helps you to justify the choice of project topic, what gap in the literature it seeks to fill, and identification of the project's beneficiaries (academic or industrial).**

Introduction - Importance

- **This leads into the rationale behind the research, revealing whether it is building upon previous research, looking at something that everybody else has overlooked, or improving upon a previous research project that delivered unclear results.**
- **This section can then flow into how you are going to fill the gap, laying out your objectives and methodology. You are trying to predict what impact your research will have if everything works as it should, and you ultimately reject the null hypothesis.**

Introduction – Research Question

- **You research question and objectives need to be precisely stated, together with the tests that will show, at the end of the project, that they have been met (or not been met).**
- **You need also to outline your methods in broad terms to show how you planned to meet the objectives.**

Introduction - Limitations

- **The introduction is the place to highlight any weaknesses in the experiment from the start.**
- **For example, an ideal experiment should have perfectly randomized samples, but there are many good reasons why this is not always possible. As long as you warn the reader about this, so that they are aware of the shortcomings, then they can easily judge the validity of the research.**
- **This is much better than making them wait until you point it out in the discussion.**

Introduction - Assumptions

- **You should also point out any assumptions that you make about conditions during the research. You should set out your basic principles before embarking upon the experiment: any research will be built around some assumptions.**

Introduction - Structure of the Report

- **Finally, outline the structure of the report, showing how it fits together.**
- **You should briefly summarise what each main section of report presents**

Introduction- Tips

- **There are a few tips that can help you write a strong introduction, arousing interest and encouraging the reader to read the rest of your work.**
 - **Keep it Short: aprox 1-2 pages**
 - A long and rambling introduction will soon put people off and lose you marks.
 - Stick closely to your outline for the paper, and structure your introduction in a similar way.



Introduction -Tips

- **Present the problem (Motivation)**
 - Can be history of the problem ?
 - Who (Expert from state of the art) says that the problem is important ?
 - What is the current impact of the problem ?
 - What are the benefits if the problem is solved or partially solved ?
- **Previous solutions to the problem**
 - Why is not the problem fully solved ?
 - What is the fundamental difficulty with the problem ?
 - Explain the technical side of the problem (Grounded in the technical sense with reference to the stat of the art)
- **Note: Paragraph on Research Question and variables in research question should be described in the above.**



Defining a Research Problem

- **Formulating the research problem begins during the first steps of the scientific process.**
- **As an example, a literature review and a study of previous experiments, and research, might throw up some vague areas of interest.**
- **Many scientific researchers look at an area where a previous researcher generated some interesting results, but never followed up. It could be an interesting area of research, which nobody else has fully explored.**
- **A scientist may even review a successful experiment, disagree with the results, the tests used, or the methodology, and decide to refine the research process, retesting the hypothesis.**
- **This is called the conceptual definition, and is an overall view of the problem. A science report will generally begin with an overview of the previous research and real-world observations. The researcher will then state how this led to defining a research problem.**

Defining a Research Question

- **What is a research question:**
 - A clear, focused and actionable proposition that guides all areas of the subsequent study from literature review, methodology, analysis and reporting.
- **The research question begins with a *research problem*, an issue someone would like to know more about or a situation that needs to be changed or addressed, such as:**
 - Conditions that could be improved
 - Difficulties/problems that need to be eliminated
 - Questions seeking answers

Defining a Research Question

- ***A research problem* leads to a research question**
- **Questions should in some way. . .**
 - Be worth investigating
 - Contribute knowledge & value to the field
 - Improve some condition
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- **Characteristics of a good research question:**
 - The question is feasible
 - The question is clear
 - The question is actionable
 - The question is significant
 - The question is ethical

Defining a Research Question

- **Use the Why-What-How method to define a research question:**
- **Why is this an important topic or an area worth investigating? (evidence from the literature)**
- **What will you do answer the research question? (a data mining experiment using classification/regression/clustering)**
- **How – what methods will you use (Naïve Bayes/Decision Tree/Support Vector Machine)**

An example Research Question

- ***Why is this important?:***
 - Diagnostic systems such as Echocardiography (ECG) to identify stages of cardiovascular diseases (CVDs) require highly skilled physicians to evaluate complex combinations of clinical and pathological data. There is currently a skills shortage in this area and the work cannot be performed by less skilled personnel as this would result in a large volume of medical error.
- ***What can be done to address the problem?:***
 - Data mining methodologies such as classification can be applied to large medical datasets to extract insights that aid less skilled healthcare professionals in the diagnosis of cardiovascular diseases.
- ***How will I carry out the work?:***
 - A hybrid model of feature selection and classification algorithms will be developed to select those features most pertinent to CVD diagnosis and to classify patients into one of four stages associated with CVD. Filter and wrapper methods will be applied to select features, and decision tree, Naive Bayes and Bayes Net classifiers will be developed to classify CVD stages. Hybrid models will be developed using combinations of feature selection and classification algorithms. The performance of the models will be evaluated in terms of accuracy, sensitivity and specificity.
- **RQ: Can a hybrid data mining model, combining feature selection and classification be used to accurately identify the stages of CVD?**

Introduction



- **Background - Announce the problem [Broad summaries of state of the art]**
- **Motivation – justify why it needs to be studied**
- **Describe variables or factors that affect the outcome**
- **Research Question and research objectives**
 - “The research question posed in this research [investigates/explores]
 - “To address the research question, the following specific sets of research objectives were derived:
 - 1 Investigate the state of the art broadly around...
 - 2 Design a ...
 - 3 Implement a ...
 - 4 Evaluate a ...”
- **Contribution**
 - “The major contribution of this research is a “
- **Structure of Paper**
- **Length: ~1-2 page**